Lecture6

基于低代码平台的信息服务开发



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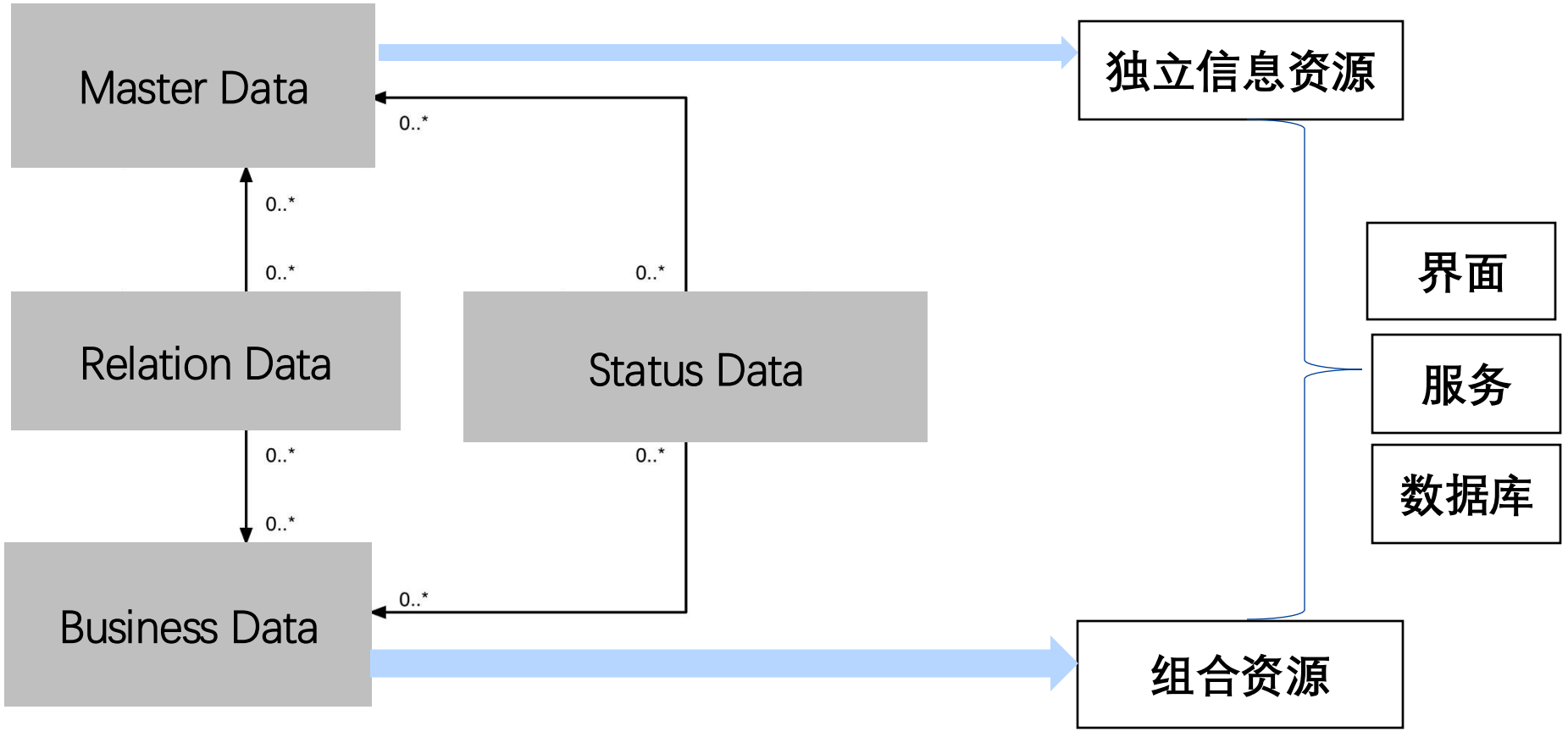
1 信息服务概述

按照时间稳定性以及应用的特点，把数据分类为：

- 主数据：主要是相对稳定、可以在公共范围内重用的基础数据，可独立存储并单独存在，主要的操作可以分为只读不写，以及单独的增删权限等。
- 事务数据：是大量生产及重复处理的业务数据。
- 关联数据：连接业务数据和主数据的中间数据，和数据处理过程相关。
- 状态数据：面向流程类应用的信息控制所建立的数据，主要体现为主数据或者事务数据的动态状态变化。
- 元数据：管理数据的数据。
- …

数据分类和模型映射

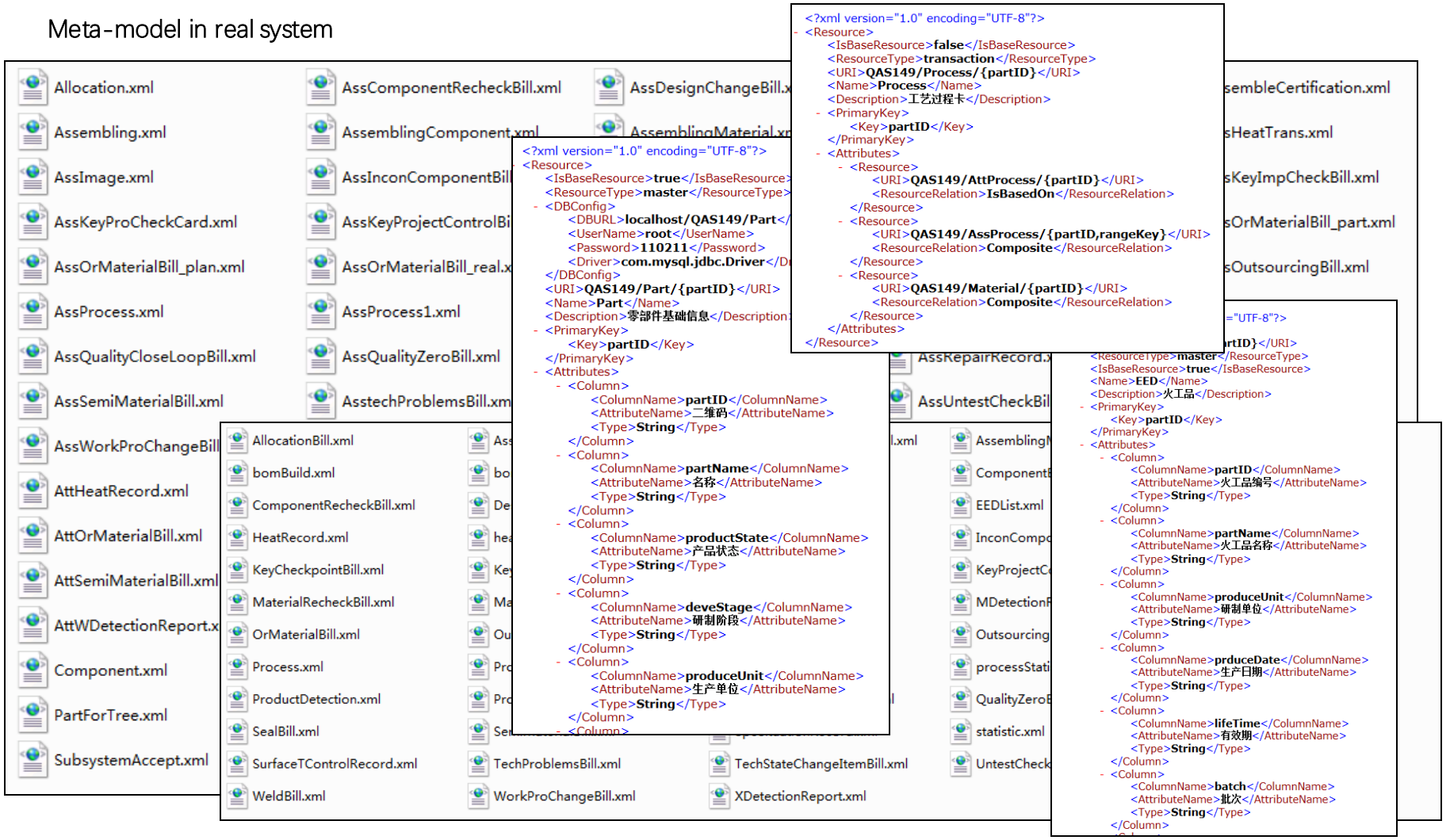
按照类型对数据建模，往往也涉及到各类数据的建模和关联：



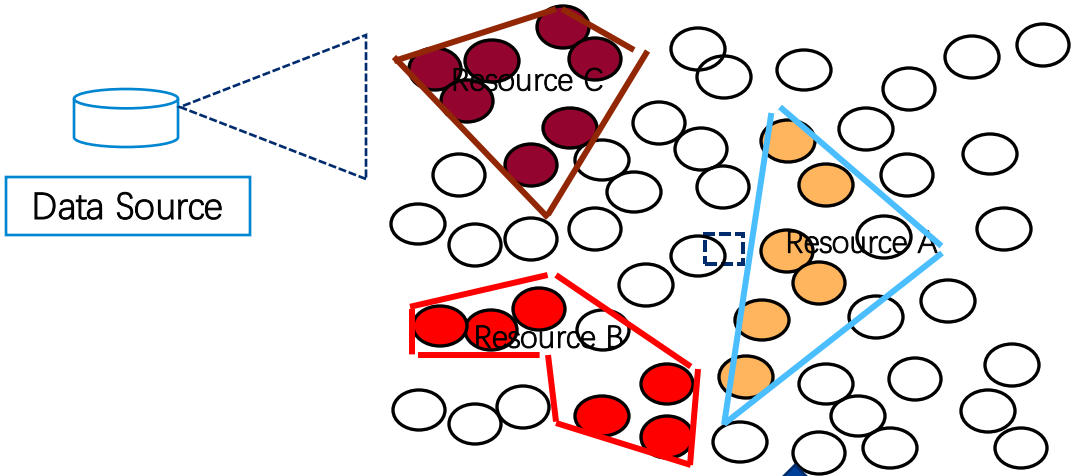
元数据

在企业中，有大量的数据实体，以及实体之间的复杂关系。

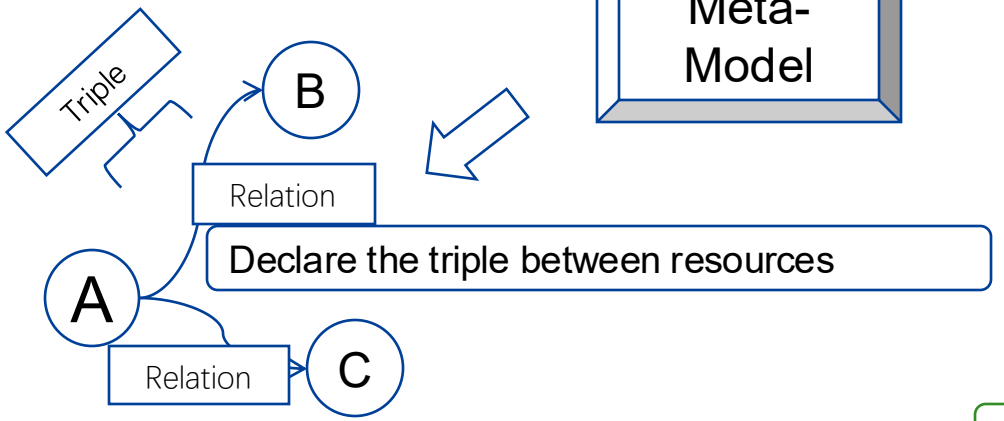
Meta-model in real system



元数据

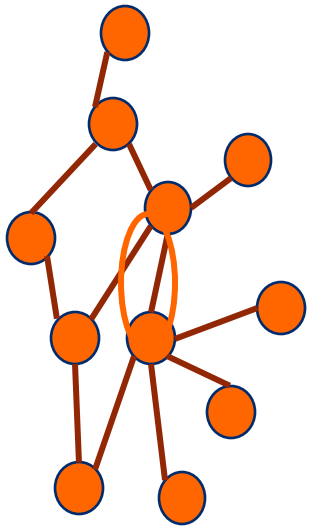


Distinguish and identify the resource clearly



Columns
Attributes
Triples
Constrains

Owl Ontology



Bridging the resource with owl-ontology

2 数据驱动的信息服务开发模式

数据驱动的软件开发模式是面向以数据为中心的系统的一类重要的开发方法，主要要点包括

- 信息资源构建：以资源表示的信息实体、资源持久性；
- 信息资源配置：
- 信息资源的服务化：
- 运行时机制：资源状态、资源锁等

（1）资源构造

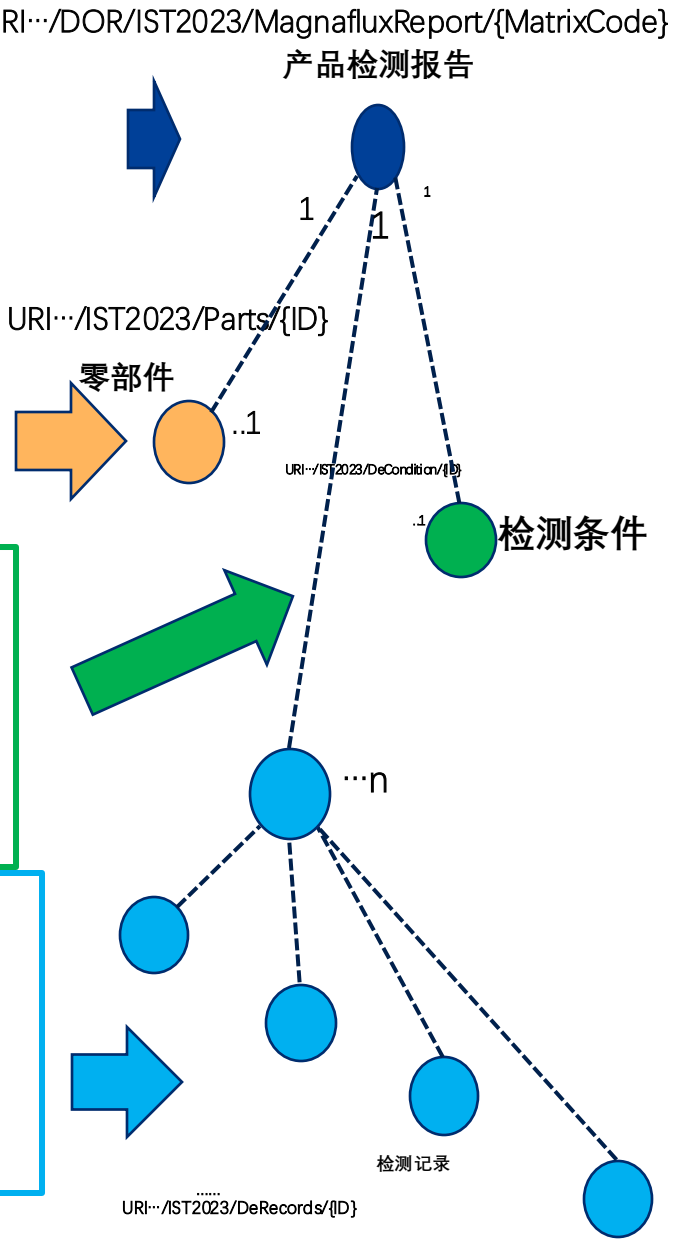
产品检测报告

送检单位：_____ 收到日期：_____ 报告日期：_____

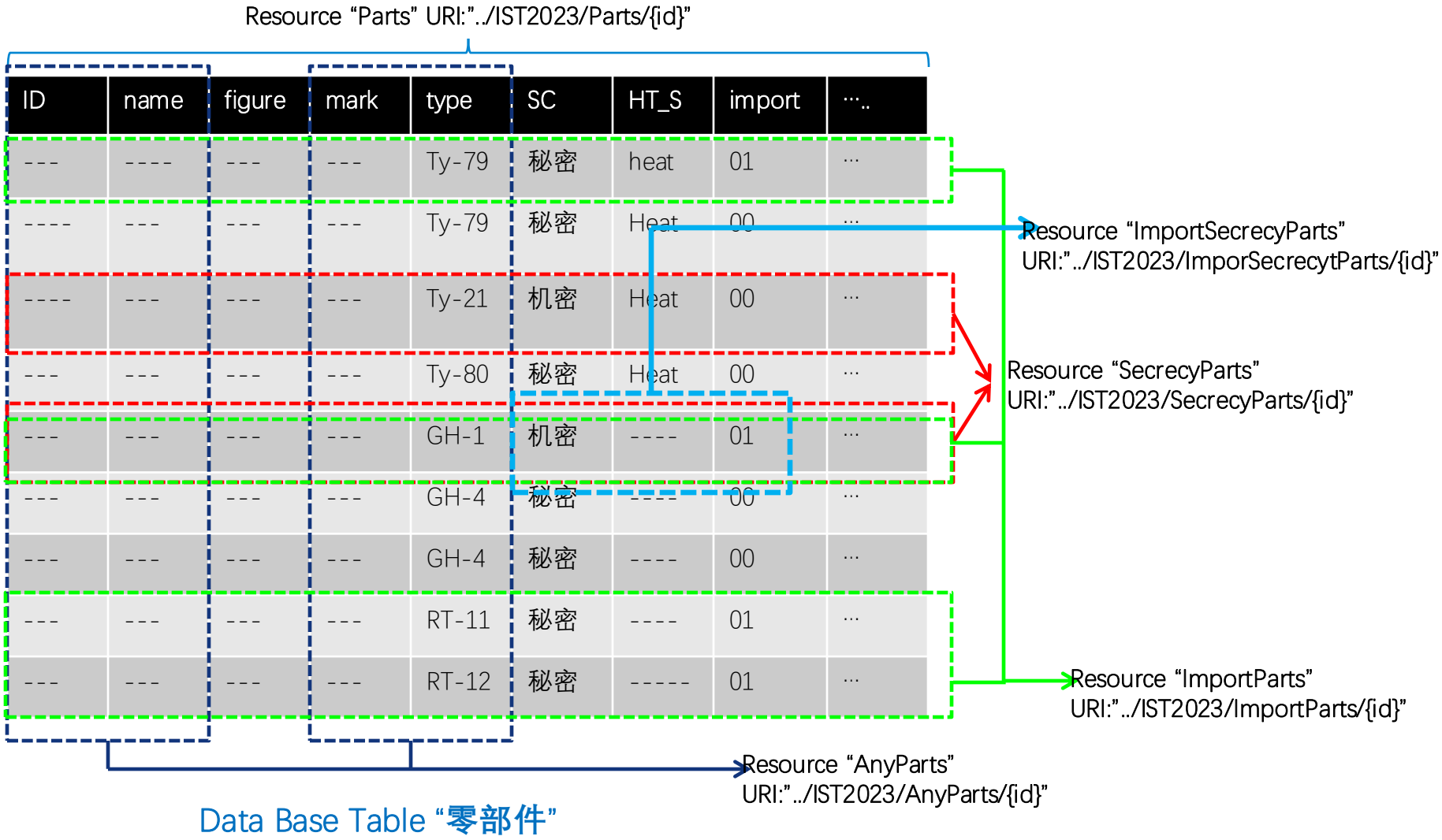
……

送检零件	名称：_____	热处理状态：_____
	图号：_____	探测面状态：_____
	牌号：_____	检验部位：_____
	工作令：_____	工艺过程卡：_____
探测条件	检验方法：_____	探头频率：_____
	仪器：_____	耦合剂：_____
	探头类型：_____	灵敏度：_____
	晶片尺寸：_____	…….

探伤记录	序号	磁粉种类	磁粉粒度	磁悬浮液	磁化方向	退磁方法	磁化方法	…….
	01	荧光	200目	17%	周向	交流	穿棒	…….
	02	荧光	300目	17%	复合	交流	夹持	…….
	N…	……	……	……	……	…….	…….	…….

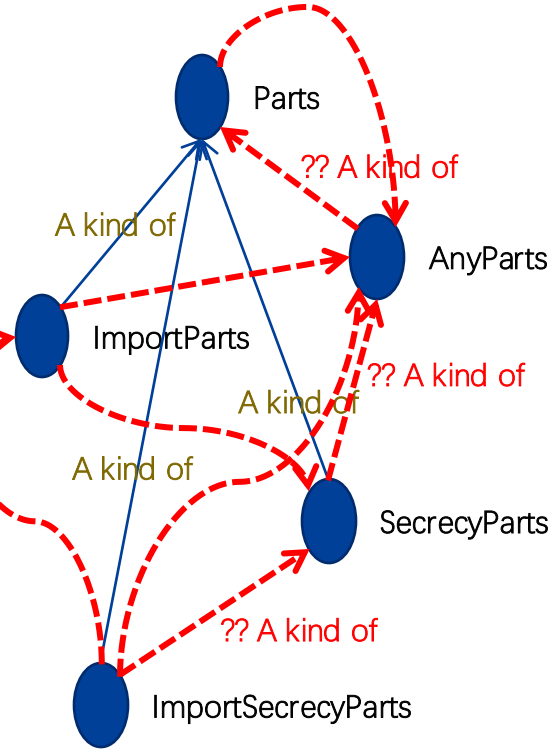


(2) 资源持久化


Resource “Parts” URI:”../IST2023/Parts/{id}”
ID
name
figure
mark
type
SC
HT_S
import
…..
Ty-79 秘密 heat 01 …
Ty-79 秘密 Heat 00 …
Ty-21 机密 Heat 00 …
Ty-80 秘密 Heat 00 …
GH-1 机密 ---- 01 …
GH-4 秘密 ---- 00 …
GH-4 秘密 ---- 00 …
RT-11 秘密 ---- 01 …
RT-12 秘密 ----- 01 …
Resource “ImportSecrecyParts”
URI:”../IST2023/ImporSecrecytParts/{id}”
Resource “SecrecyParts”
URI:”../IST2023/SecrecyParts/{id}”
Resource “ImportParts”
URI:”../IST2023/ImportParts/{id}”
Resource “AnyParts”
URI:”../IST2023/AnyParts/{id}”
Data Base Table “零部件”

Resource Definition and Persistence

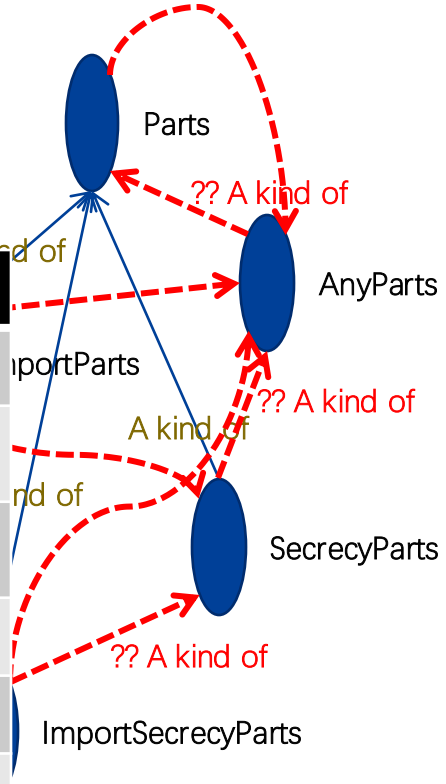
ID	name	figure	mark	type	SC	HT_S	import	….
---	----	---	---	Ty-79	秘密	heat	01	…
----	---	---	---	Ty-79	秘密	Heat	00	…
----	---	---	---	Ty-21	机密	Heat	00	…
---	---	---	---	Ty-80	秘密	Heat	00	…
---	---	---	---	GH-1	机密	----	01	…
---	---	---	---	GH-4	秘密	----	00	…
---	---	---	---	GH-4	秘密	----	00	…
---	---	---	---	RT-11	秘密	----	01	…
---	---	---	---	RT-12	秘密	-----	01	…



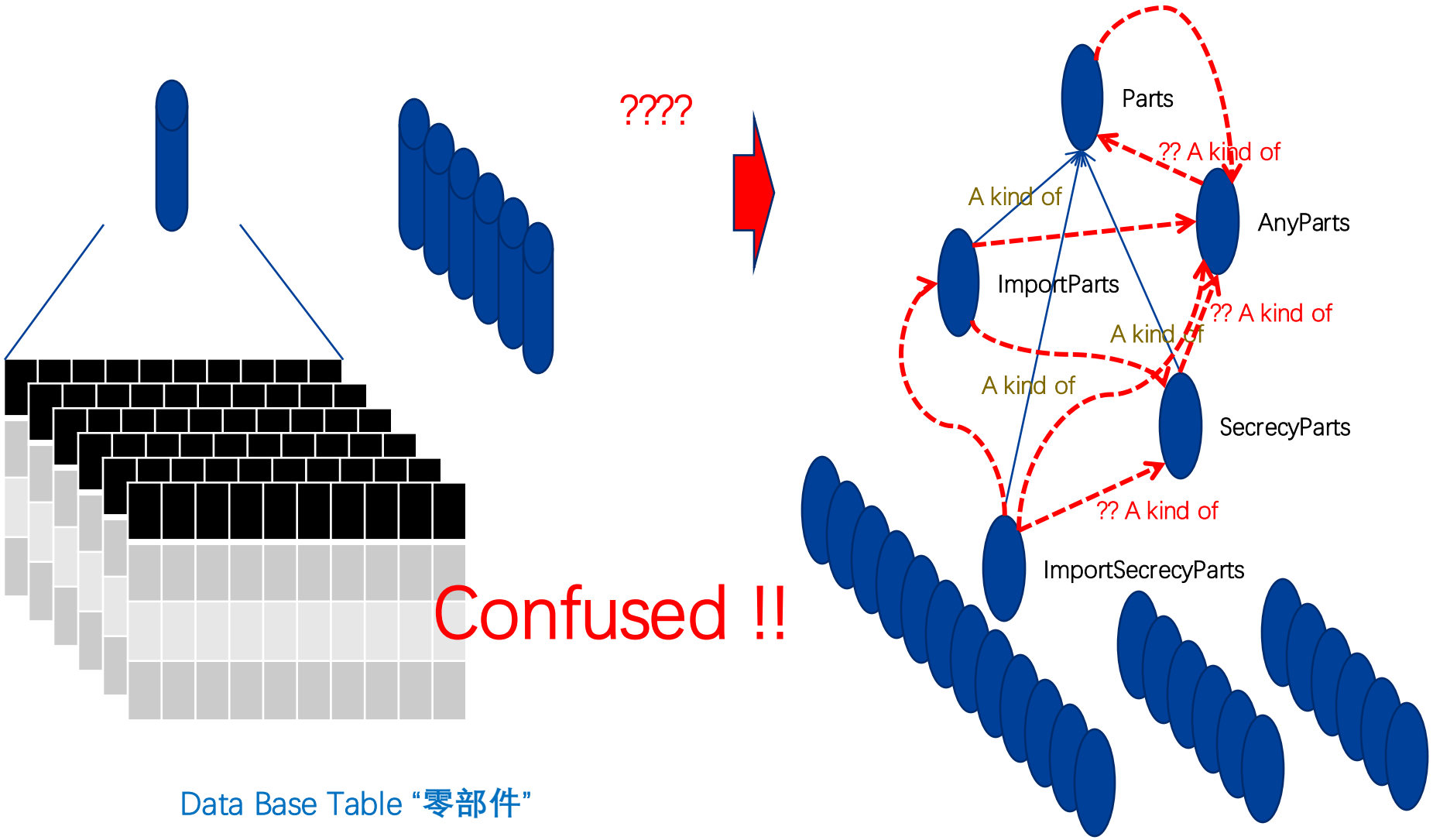
Data Base Table “零部件”

Resource Definition and Persistence

ID	name	figure	mark	type	SC	HT_S	import	···..
---	----	---	---	Ty-79	秘密	heat	01	···
----	---	---	---	Ty-79	秘密	Heat	00	···
----	---	---	---	Ty-21	机密	Heat	00	···
---	---	---	---	Ty-80	秘密	Heat	00	···
---	---	---	---	GH-1	机密	----	01	···
---	---	---	---	GH-4	秘密	----	00	···
---	---	---	---	GH-4	秘密	----	00	···
---	---	---	---	RT-11	秘密	----	01	···
---	---	---	---	RT-12	秘密	-----	01	···



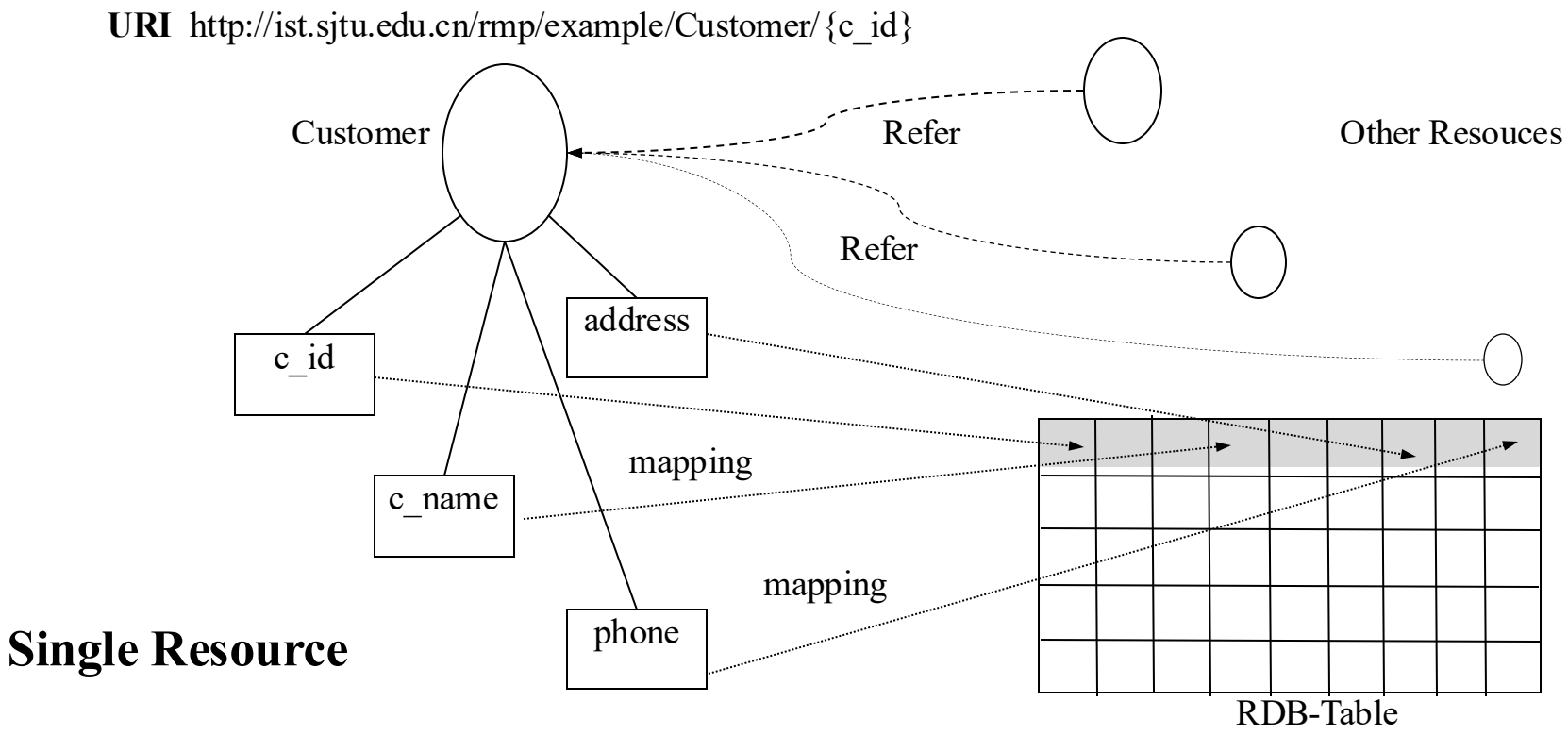
Resource Definition and Persistence



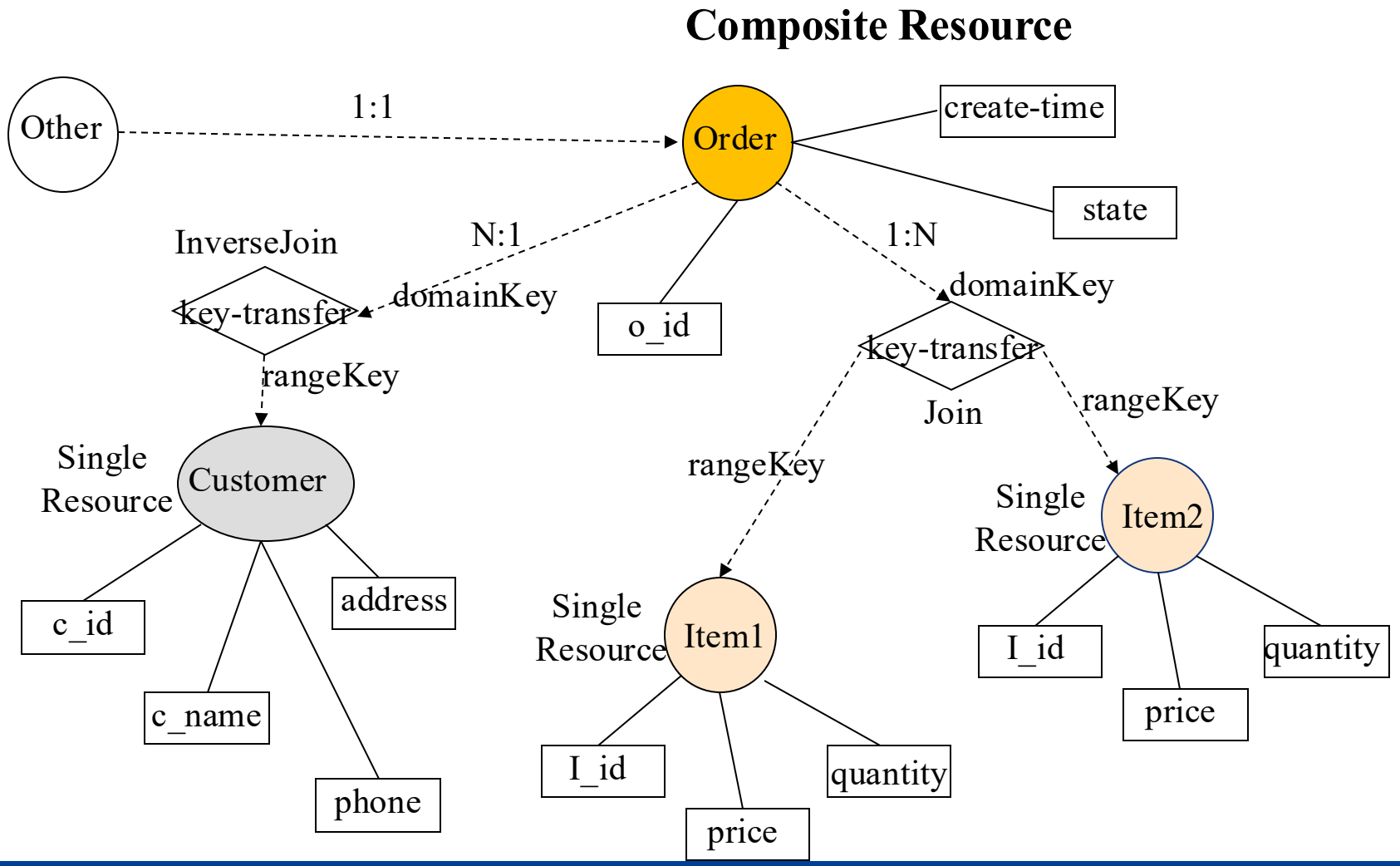
Data Base Table “零部件”

（3）资源配置

- 资源实体模型主要分为单一（独立）资源和复合资源。



- 复合资源可以由单一资源配置而来。



Tips for resources configuration

- Modeling master data as single/independent resource.
  - 变化少的，基本数据
- Modeling transaction data as composite resource.
  - 业务相关，变化频繁

- Operational Semantics is based on Resource Relation Reasoning.

（4）资源的服务封装

➢REST: represent resources as services

Car with services

None-resource service: RPC, SOAP services etc.





Emergency!

Function: start(Car yourCar, Key yourkey)

Function: park(Car yourCar, Location place)

Function: stopEngine(Car yourCar, Key yourkey)

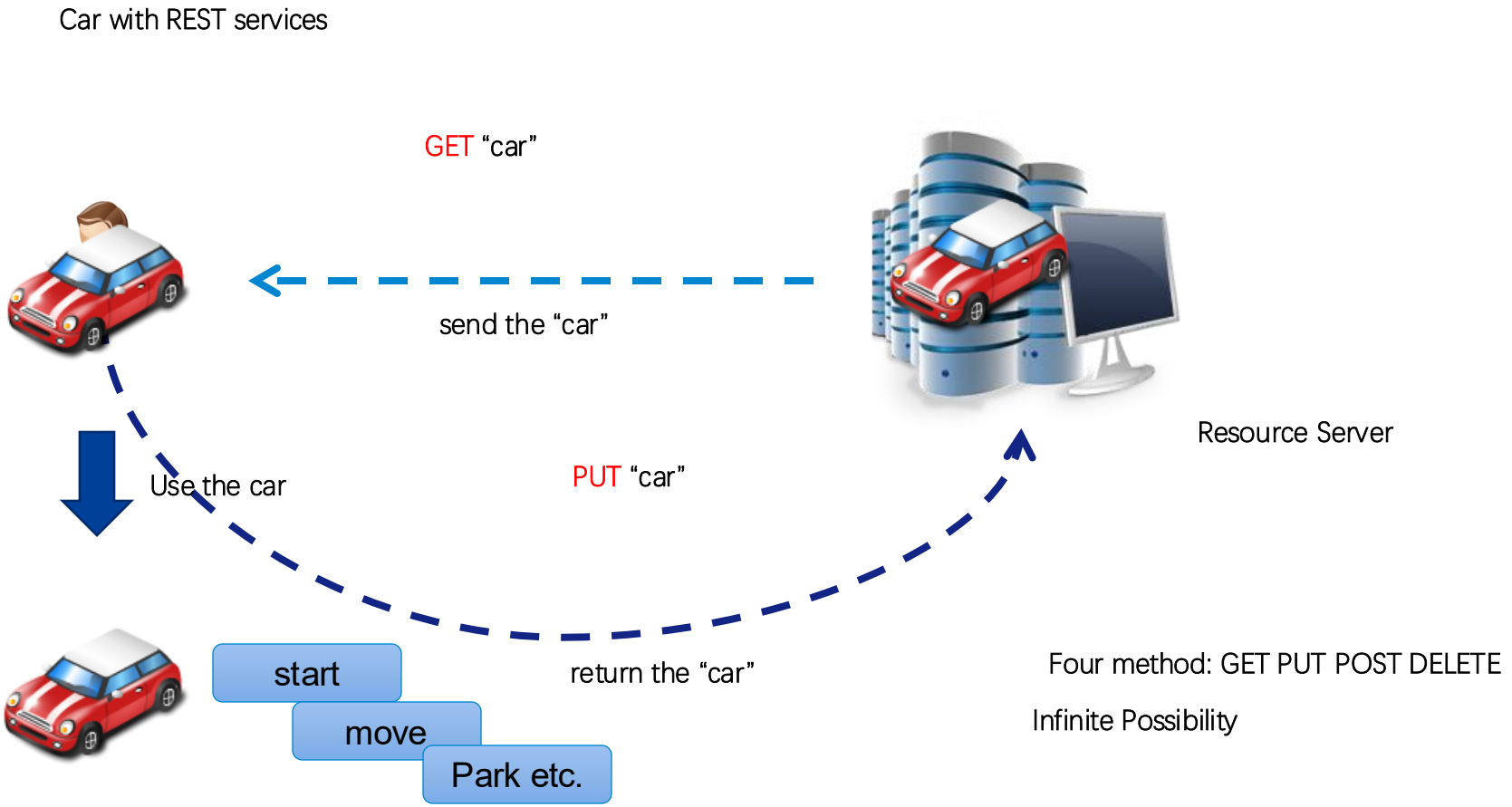
Function: breakTheWindow(Car yourCar, CellPhon



Server Response: NO SUCH METHOD SUPPORT !

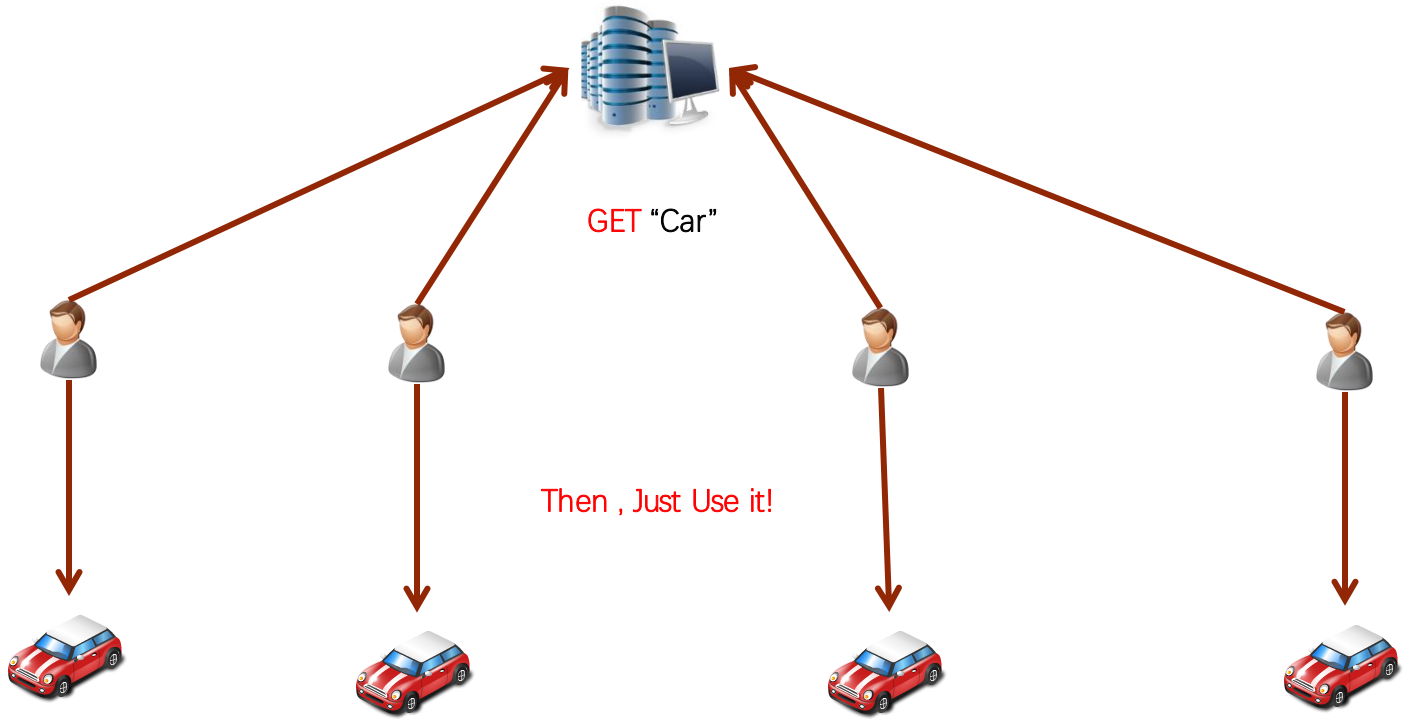
将信息资源封装为数据服务

➢represent resources as services



➢represent resources as services

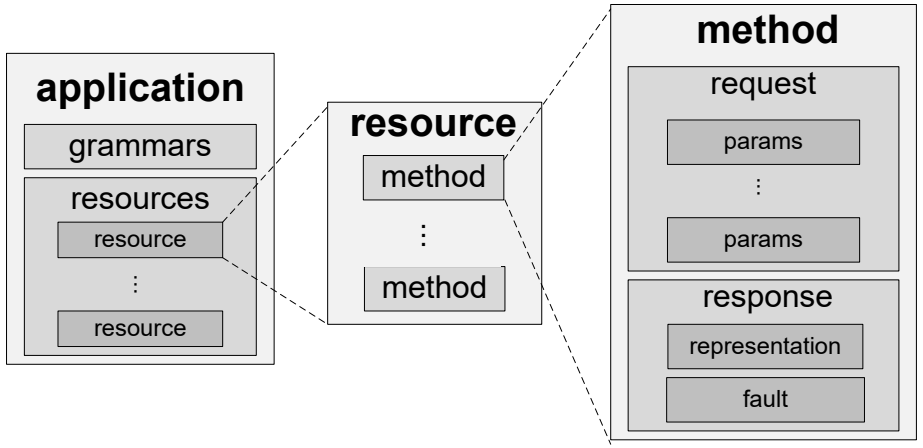
Car with REST services



RESTful service generation

WADL and WSDL 2.0 are used to represent RESTful service.

Based on resource model, RESTful service description could be generated automatically.




```
<?xml version="1.0" encoding="UTF-8" ?>
- <application xmlns:xsi="http://wss.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://weblogs.java.net/blog/mhadley/wadl.xsd">
  - <resources base="http://example.org/customer">
    - <resource path="order">
      - <method name="PUT">
        - <request>
            <param name="customerid" type="xsd:string" required="true" />
            <param name="productids" required="true" />
          </request>
        - <response>
            <representation mediaType="application/xml" element="order" />
          </response>
        </method>
      - <method name="GET">
        - <request>
            <param name="orderid" type="xsd:string" required="true" />
          </request>
        - <response>
            <representation mediaType="application/xml" element="order" />
          </response>
        </method>
      </resource>
    - <resource path="order/availableOrder">
      - <method name="DELETE" id="cancelAvailableOrder">
        - <request>
            <param name="orderid" type="xsd:string" required="true" />
          </request>
        - <response>
            <representation mediaType="application/xml" element="order" />
          </response>
        </method>
      </resource>
    - <resource path="order/payedOrder">
      - <method name="POST" id="createPayedOrder">
        - <request>
            <param name="orderid" type="xsd:string" required="true" />
          </request>
        - <response>
            <representation mediaType="application/xml" element="order" />
          </response>
        </method>
      </resource>
    - <resource path="order/sentOrder">
      - <method name="POST" id="send">
        - <request>
            <param name="orderid" type="xsd:string" required="true" />
          </request>
        - <response>
            <representation mediaType="application/xml" element="order" />
          </response>
        </method>
      </resource>
    </resources>
  </application>
```

(5) 运行机制

Runtime mechanism is for system control logic:

(1) Resource states

- State is also a kind of Resource
- State-Resource has no instances
- State-Resource should describe in ontology
- All State-Resource can be organized in a space
- Every resource just refer the State-Resource
- State-Space provides the ways and user decides the one (nondeterministic)

(2) Resource locks

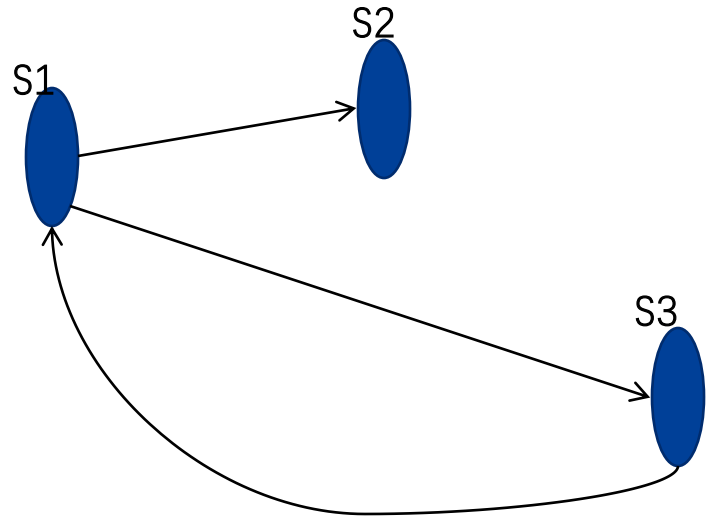
(3) Access Authority

Resource states

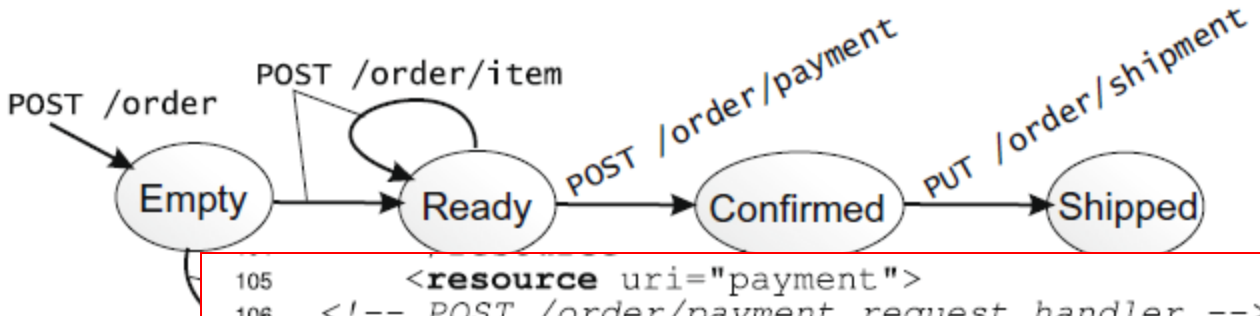
Finite-state machines (FSMs)

- are a mathematical formalism for describing processes with a finite number of possible states and sequential state transitions.

R	S1	S2	S3
S1	1	1	1
S2	0	1	0
S3	1	0	1



Resource states



BEPL and RESTful

```
105       <resource uri="payment">
106   <!-- POST /order/payment request handler -->
107         <onPost>
108          <sequence>
109           <variable name="payment_response"/>
110           <assign>
111           amount = computeTotalOrderAmount($items);
112           payment = $request.payment;
113           payment.amount = $amount;
114           </assign>
115           <post url="https://www.visa.com/payment"
116                 request="payment" response="payment_response">
117            <catch code="200">
118             <sequence>
119              <assign>
120               state = "Confirmed";
121               payment.id = payment_response.confirmation;
122              </assign>
123              <respond code="200">
124               Payment has been accepted: $payment_response
125              </respond>
126             </sequence>
127            </catch>
```

Resource locks

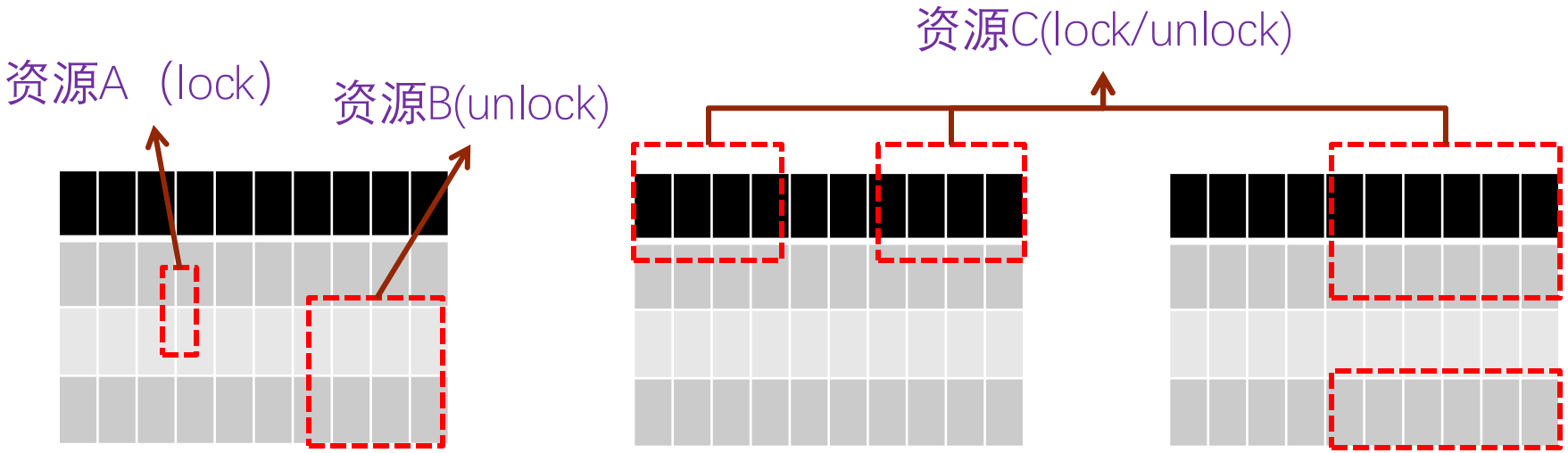
传统意义的数据库锁（定粒度锁）

数据库锁（schema）>数据表锁（Table）>记录锁（Record）

数据库锁的操作由具体用户根据实际应用需求制定,需要用户对数据表及关联数据有清晰的认识

RMP平台资源锁（变粒度锁）

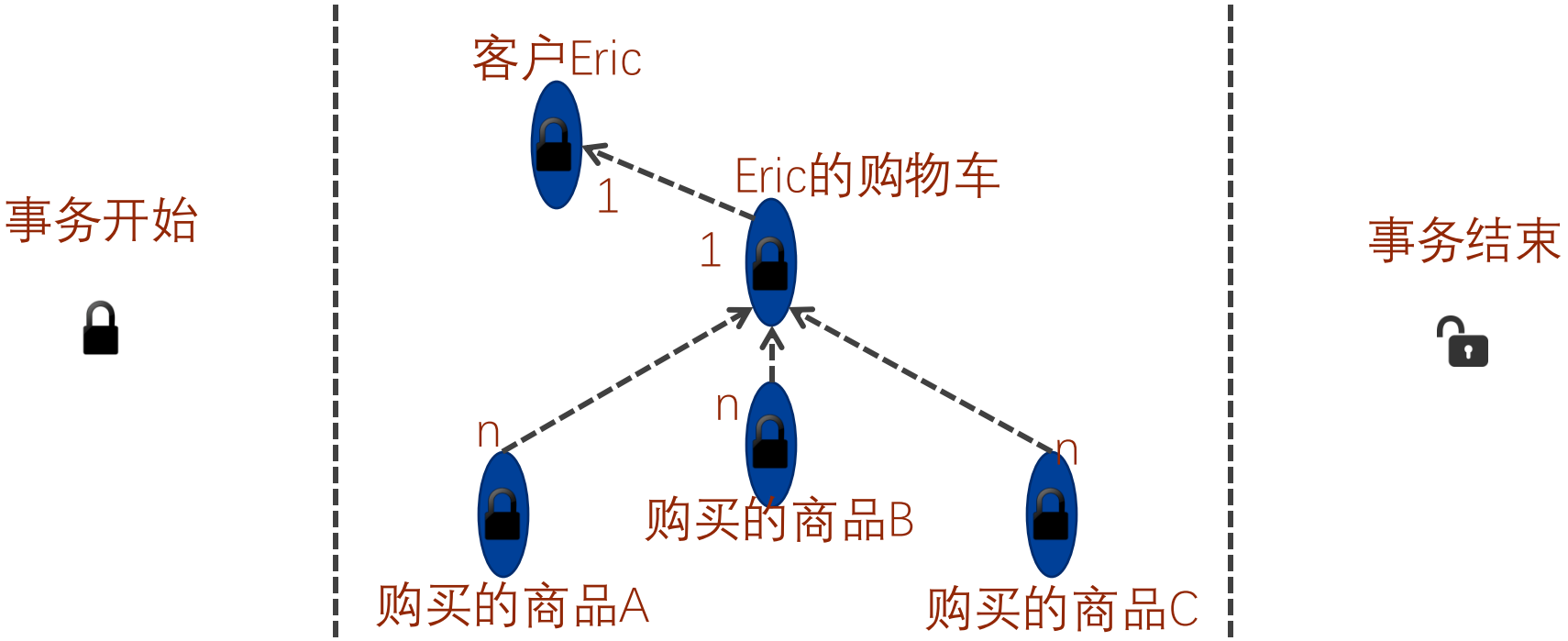
资源锁的操作由RMP资源事务层控制，对用户透明。



资源锁操作

通用的：先到先得，短（长）事务优先，高优先级优先，等等。死锁检查、死锁预防、两阶提交等。

业务例子：客户Eric挑选商品加入购物车并提交。（写锁）



Access Authority

➢资源权限控制

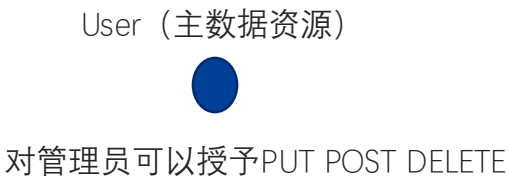
所有资源至多拥有4个权限：GET、PUT、POST、DELETE。

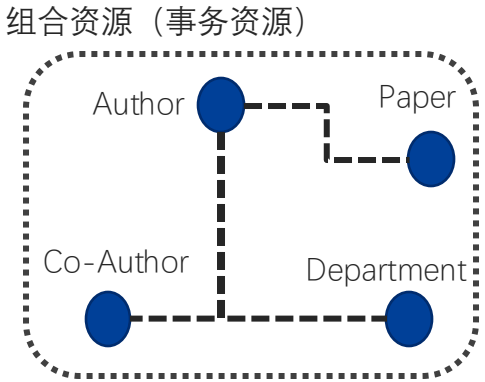
RMP平台对资源权限的控制完全基于对4种操作的变粒度组合。

➢主数据资源与事务资源

主数据资源：为其他资源或事务提供引用，一般授予GET权限

事务资源：业务执行过程中的操作对象，根据范围，授予PUT等





通常，组合资源是一种事务资源。对组合资源的PUT操作首先作用于组合内部的组合关系。

其次，如果被组合的资源是事务资源，则对组合资源的PUT操作也会影响该事务资源（级联）。

最后，如果被组合的资源是主数据资源，则对组合资源的PUT操作不会影响该资源。（不级联）

Access Authority

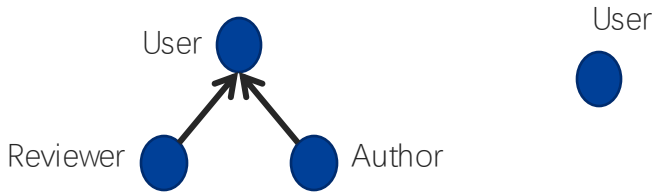
参考业务:

1.管理员可以对User资源进行任意操作

授予admin资源对User资源的GET、PUT、POST、DELETE

2.审稿人(Reviewer)只能查看作者（Author），而不能查看或修改User

（1）从资源User中抽象出资源Reviewer和Author



（2）授予资源Reviewer对Author的GET权限

3.作者只能修改自己的文章

（1）移除作者对Paper资源的PUT权限

（2）组合Author与Paper形成组合资源

（3）授予作者对该组合的PUT权限



4.客户想要修改商品信息

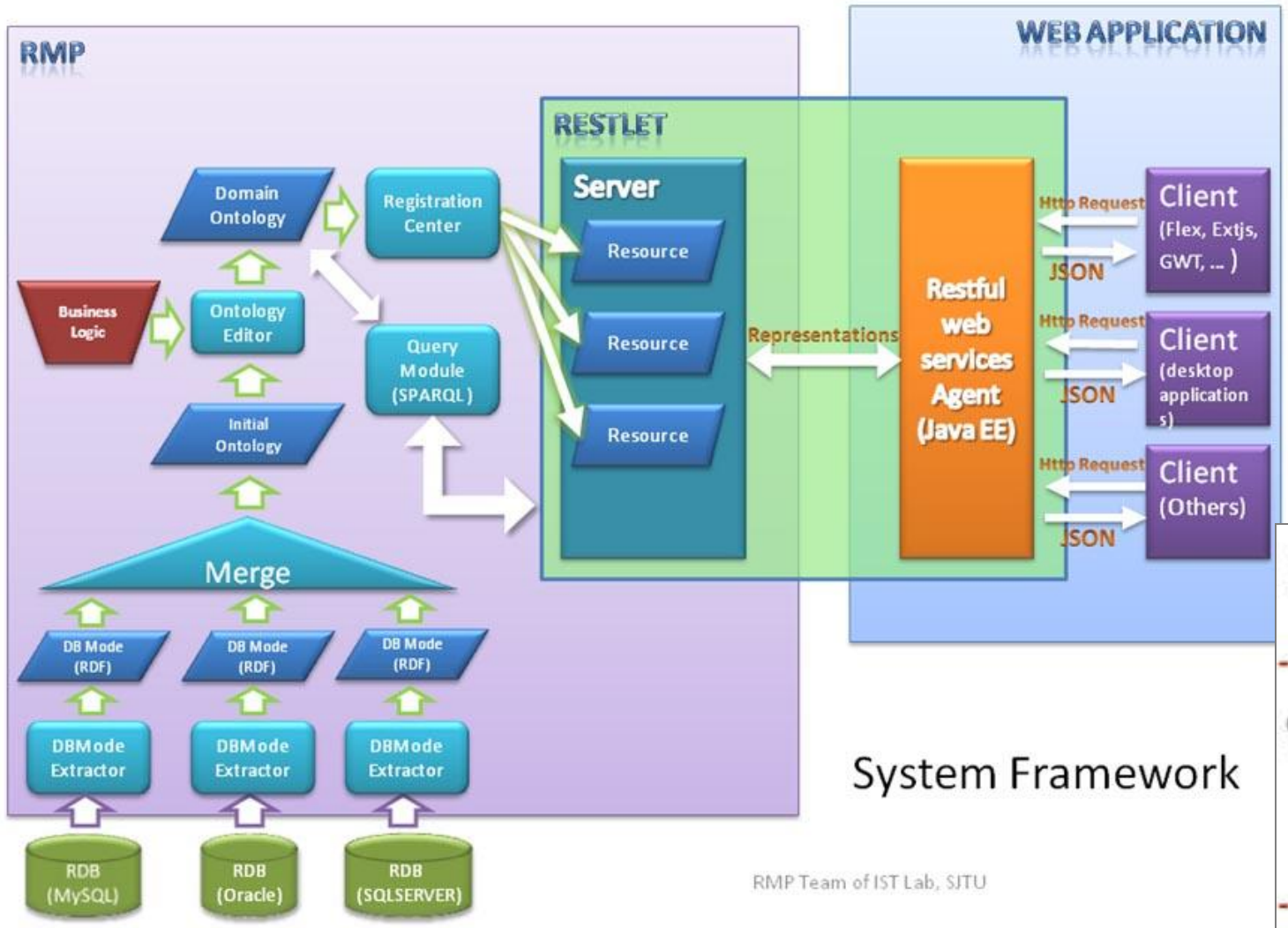
商品被设置为主数据资源，客户不能修改商品信息（即使该商品已被加入该客户的购物车）。客户只能修改购买商品的数量、发货地址等。

3 基于 RMP 平台的信息服务开发

资源管理平台(RMP)，基于本体建模的跨系统企业资源集成管理平台，是IST开发的数据驱动的Web应用系统开发平台。

- 平台针对网络环境中的企业信息系统，业务人员通过使用系统可以灵活的在基于语义的环境下建立业务资源，通过本体管理业务资源模型，并通过Service访问调用资源完成业务需求；
- 以往信息系统中的信息实体只通过XML直接反映数据库中的关系型结构，而本系统通过对资源的语义封装和使用本体来管理资源，实现业务人员对数据组织形式的灵活选择，根据不同的业务需求和变更来订制不同的资源形式，进行调用，而无需修改数据库中的原有结构。

系统架构(RMP1.2.X)



System Framework

RMP Team of IST Lab, SJTU



基于RMP的系统开发主要要素

- 业务信息资源
  - 独立信息资源构造
  - 复合信息资源配置
- 数据库生成
  - 资源模型与数据库的映射及对应
- 资源服务的转换
  - 信息资源生成RESTful服务
- RESTful服务应用
  - 基于MVC模式的完整系统构造（权限）
  - RESTful服务与多种终端（GUI,JS,手机端）的绑定



4 数据驱动的信息服务实例

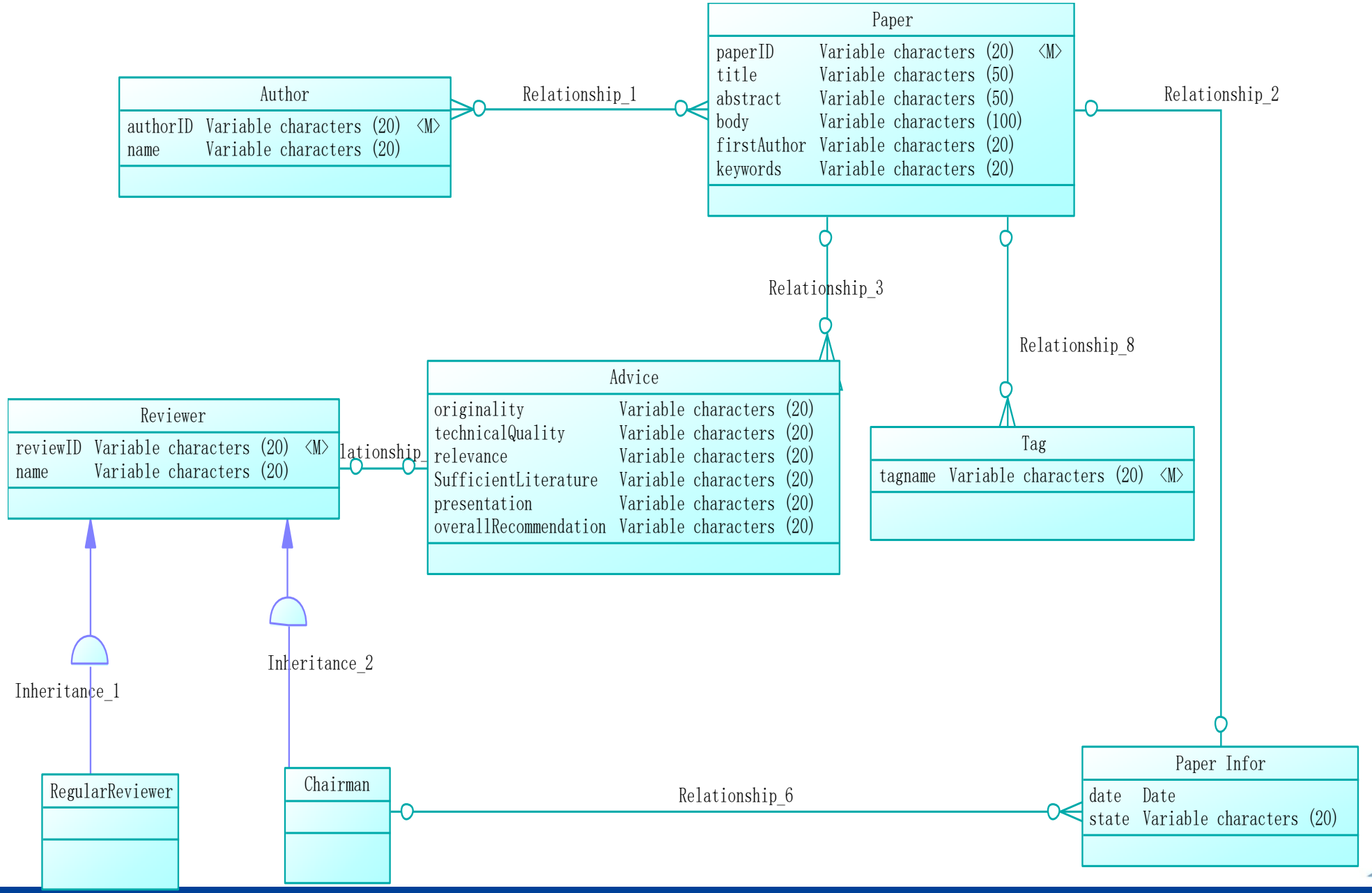
- 国际会议规模，以一个中小型ICWS为例：
  - 600多人参会，6个联合会议一起召开。
  - Research track 论文100多篇，录用率19%，8个Track；
  - 投稿800篇，审稿800*（3-5）次，PC Chair讨论以录用论文200篇次，出版归档100*3次。

- 围绕论文（主数据对象）为核心的操作处理，状态改变过程。

主要功能场景

➢ 投稿流程	作者
➢ 审稿流程	审稿人
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A reference data model



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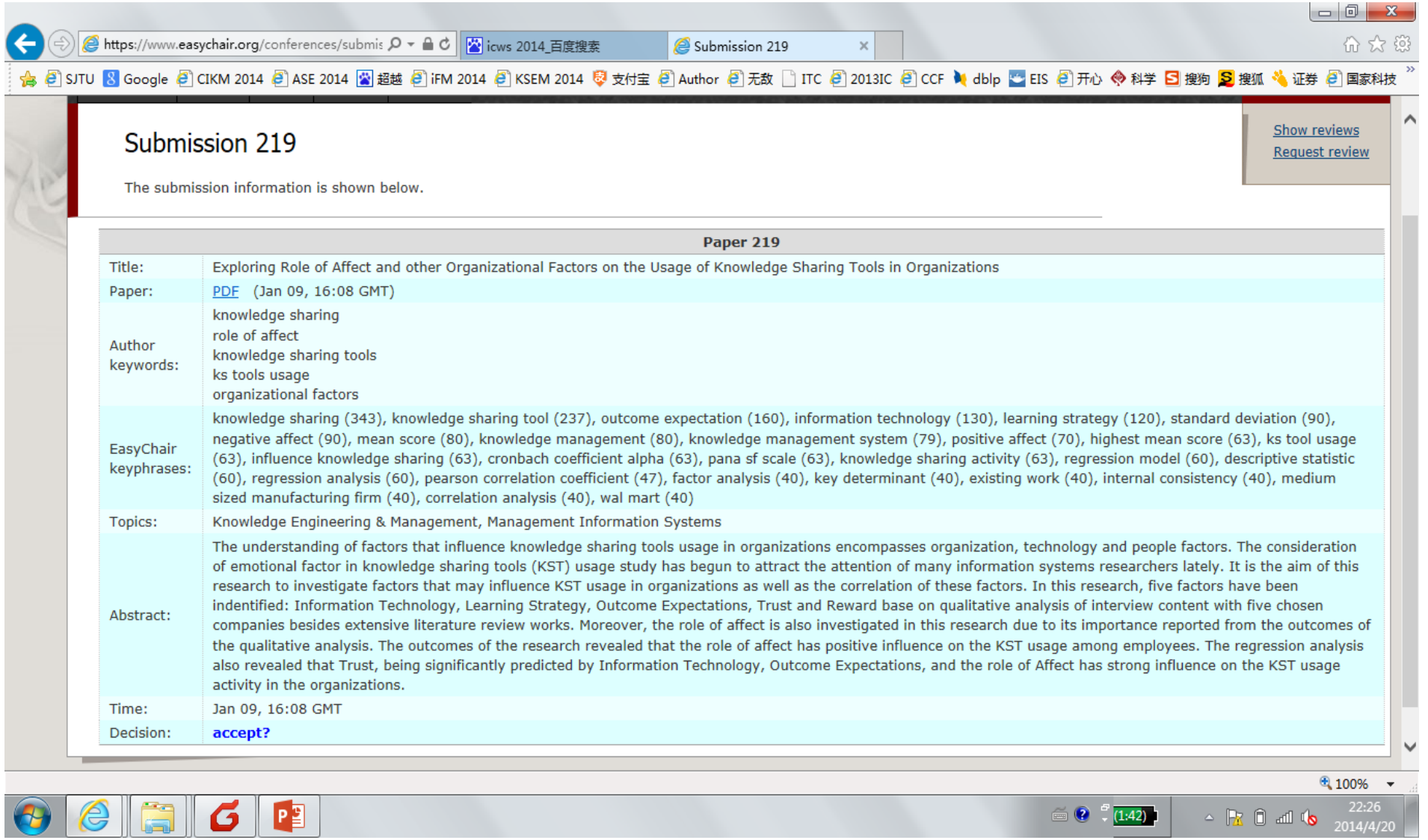
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Submission 219

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The submission information is shown below.

Paper 219	
Title:	Exploring Role of Affect and other Organizational Factors on the Usage of Knowledge Sharing Tools in Organizations
Paper:	PDF (Jan 09, 16:08 GMT)
Author keywords:	knowledge sharing role of affect knowledge sharing tools ks tools usage organizational factors
EasyChair keyphrases:	knowledge sharing (343), knowledge sharing tool (237), outcome expectation (160), information technology (130), learning strategy (120), standard deviation (90), negative affect (90), mean score (80), knowledge management (80), knowledge management system (79), positive affect (70), highest mean score (63), ks tool usage (63), influence knowledge sharing (63), cronbach coefficient alpha (63), pana sf scale (63), knowledge sharing activity (63), regression model (60), descriptive statistic (60), regression analysis (60), pearson correlation coefficient (47), factor analysis (40), key determinant (40), existing work (40), internal consistency (40), medium sized manufacturing firm (40), correlation analysis (40), wal mart (40)
Topics:	Knowledge Engineering & Management, Management Information Systems
Abstract:	The understanding of factors that influence knowledge sharing tools usage in organizations encompasses organization, technology and people factors. The consideration of emotional factor in knowledge sharing tools (KST) usage study has begun to attract the attention of many information systems researchers lately. It is the aim of this research to investigate factors that may influence KST usage in organizations as well as the correlation of these factors. In this research, five factors have been indentified: Information Technology, Learning Strategy, Outcome Expectations, Trust and Reward base on qualitative analysis of interview content with five chosen companies besides extensive literature review works. Moreover, the role of affect is also investigated in this research due to its importance reported from the outcomes of the qualitative analysis. The outcomes of the research revealed that the role of affect has positive influence on the KST usage among employees. The regression analysis also revealed that Trust, being significantly predicted by Information Technology, Outcome Expectations, and the role of Affect has strong influence on the KST usage activity in the organizations.
Time:	Jan 09, 16:08 GMT
Decision:	accept?

100%

(1:42) 22:26 2014/4/20

34

A interface of PC chair

PIC2014 List of Submissions

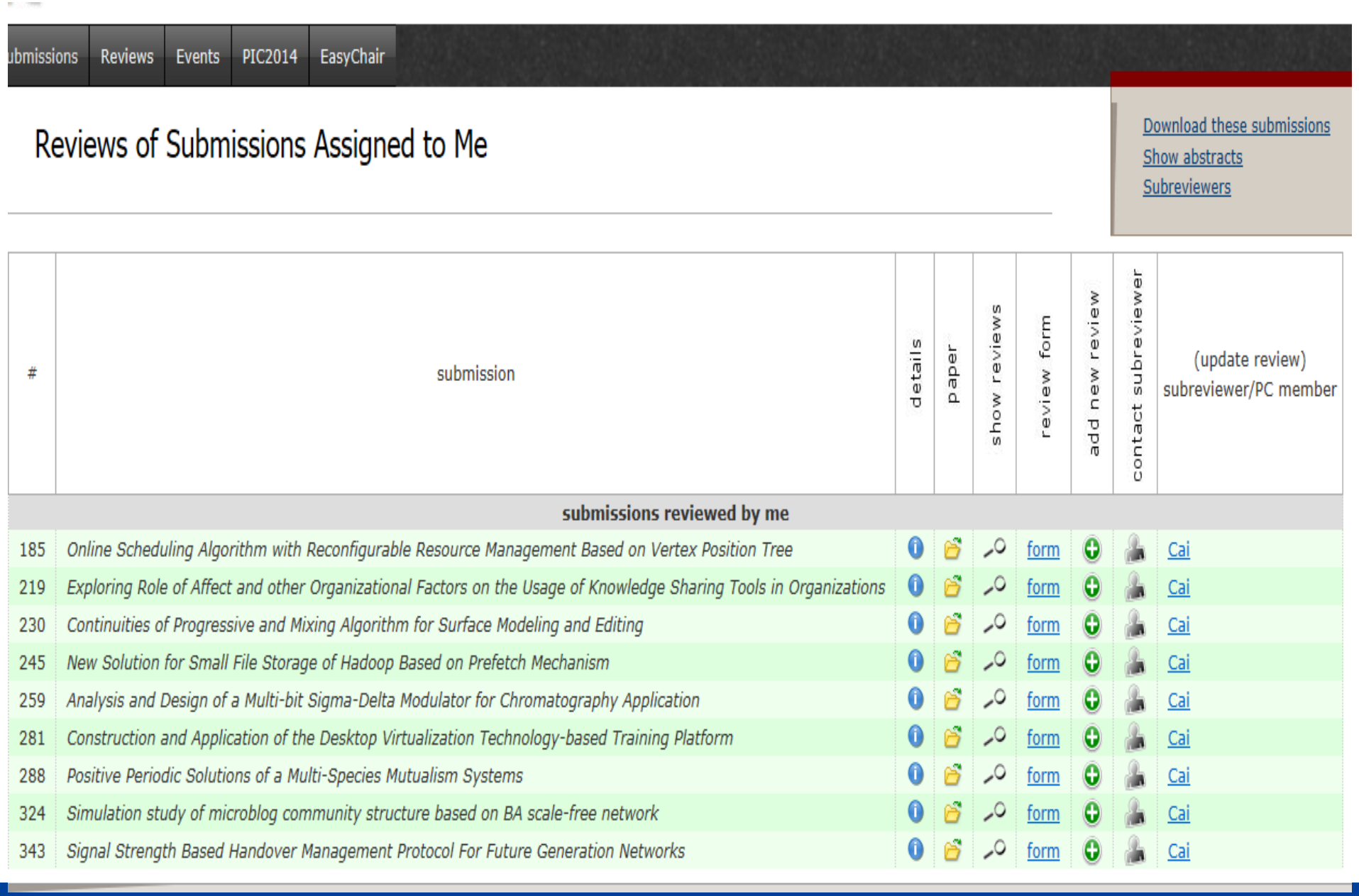
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The time in the table is the last modification time.

#	title	rm ation	m ission	time
13	3D Fountain Simulation based on Particle System			Nov 05, 08:40
14	An Improved Approach of Data Integration Based on Differential Privacy			Apr 21, 00:47
15	PIC/MCC simulation for oxygen microwave breakdown at atmospheric conditions			Jan 15, 14:46
16	Word Sense Disambiguation Based on Improved VSM			Nov 10, 17:15
17	Parameters Estimation Accuracy Analysis for CoMP System			Mar 31, 08:58
18	Training Sequence Design for CoMP System			Mar 30, 03:47
19	Laser Based Tomographic System for Visualization and Characterization of Tissue Optical Phantoms			Nov 12, 15:43
20	The Performance Analysis on Visible Light Communications Multi-LED Parallel Superposition High Speed Transmission			Apr 15, 09:13
21	A Round Robin with Multiple Feedback Job Scheduler in Hadoop			Apr 16, 06:04
22	Design of Space Class Library for 3D Geological Modeling			Nov 14, 14:36
24	Stress Analysis of Reciprocating Pump Pipeline System in Oil Station			Nov 15, 05:04
26	Energy Conservation Scheme for WSN Based on Data Correlation Clustering			Nov 18, 04:17
27	A rapid payment confirm scheme base on QRCode barcode			Nov 18, 12:30
29	Research on Partner Selection of Industrial Cluster Enterprises Based on Knowledge Complementary			Nov 19, 02:36
30	Analysis on Auto-animato System Evolution of Wuhan shipping center			Dec 03, 06:32
31	An Operation-oriented Document Natural Language Understanding Method Based on Event Model			Mar 30, 07:17
32	A new clustering algorithm with adaptive attractor for LIDAR points			Mar 30, 02:59

Reviewers interface



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	submissions reviewed by me							
185	*Online Scheduling Algorithm with Reconfigurable Resource Management Based on Vertex Position Tree*				form			Cai
219	*Exploring Role of Affect and other Organizational Factors on the Usage of Knowledge Sharing Tools in Organizations*				form			Cai
230	*Continuities of Progressive and Mixing Algorithm for Surface Modeling and Editing*				form			Cai
245	*New Solution for Small File Storage of Hadoop Based on Prefetch Mechanism*				form			Cai
259	*Analysis and Design of a Multi-bit Sigma-Delta Modulator for Chromatography Application*				form			Cai
281	*Construction and Application of the Desktop Virtualization Technology-based Training Platform*				form			Cai
288	*Positive Periodic Solutions of a Multi-Species Mutualism Systems*				form			Cai
324	*Simulation study of microblog community structure based on BA scale-free network*				form			Cai
343	*Signal Strength Based Handover Management Protocol For Future Generation Networks*				form			Cai

Decision making function

Reviews and Comments on Paper 185

Submission details
Add comment
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Edit note

Paper: (anonymous). *Online Scheduling Algorithm with Reconfigurable Resource Management Based on Vertex Position Tree*

Current decision: **ACCEPT** (accept)

Summary of Received Reviews and Comments

Reviews superseded by other reviews are shown in the grey color in the table. All times are GMT.

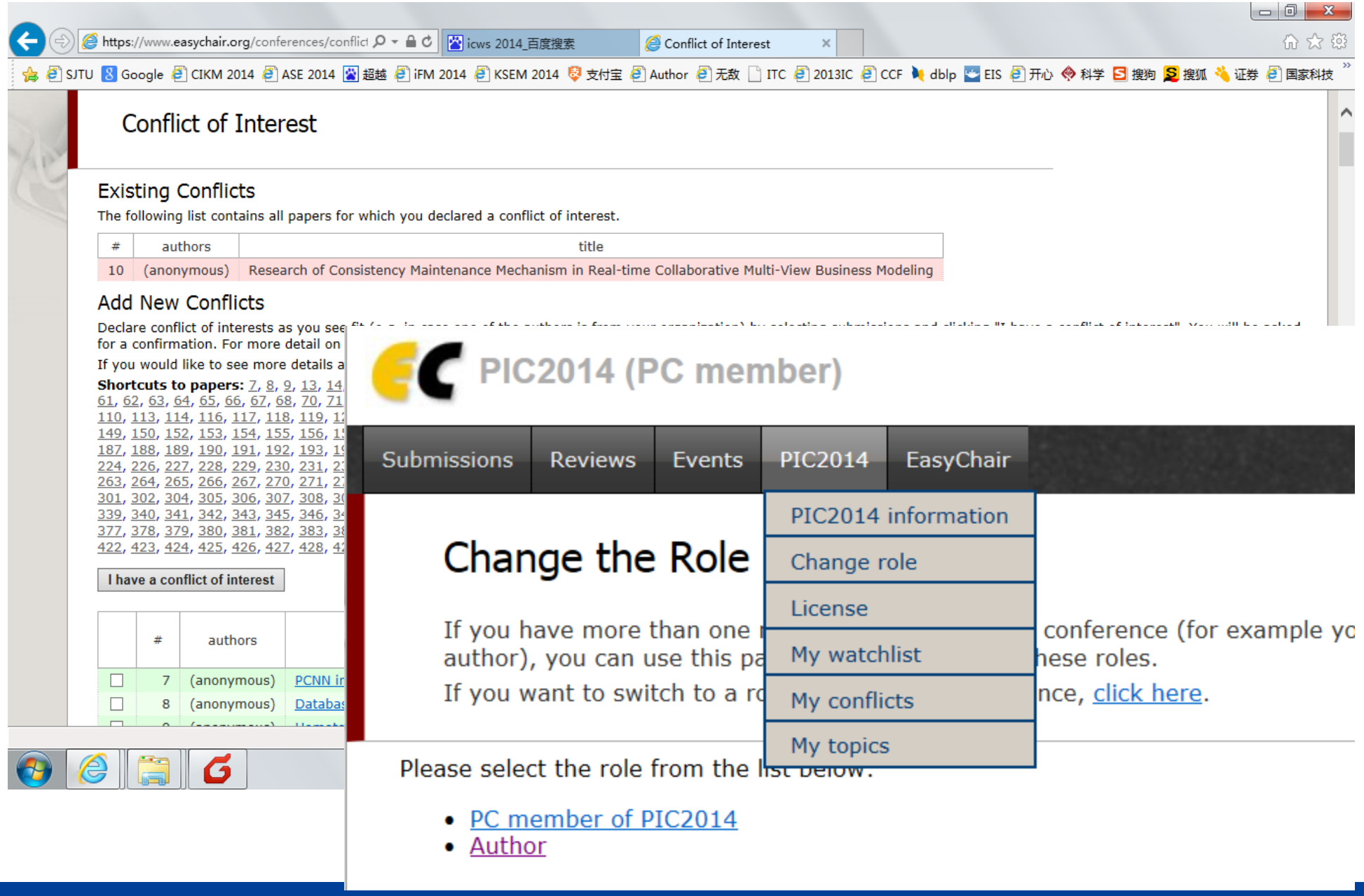
	date	PC member	subreviewer	Overall evaluation	Reviewer's confidence	
Review 1	Feb 27	Hongming Cai		**2**	**4**	(revise)
Review 2	Mar 4	Andy Connor		**2**	**2**	
Review 3	Mar 5	Shu-Ching Chen	Fausto Fleites	**2**	**3**	

Review 1	
PC member:	Hongming Cai
Overall evaluation:	**2**: (accept)
Reviewer's confidence:	**4**: (high)
Review:	In this paper, the authors introduce an online scheduling algorithm with reconfigurable resource management, in which Vertex position tree, and fragmentation shape edges are introduced. The author also use priority greedy scheduling policy for online tasks. The algorithm is useful and the paper is well organized. Hence, it can be accepted by PIC-2014. Some Weakness: there are some syntax error, e.g., "Finding the available start time and appropriate placement position for arrival tasks are the key points..." shuld be "Finding ... for arrival tasks is the key points...", so ,please check your paper carefully.
Confidential remarks for the program committee:	
Time:	Feb 27, 19:28

Review 2	
PC member:	Andy Connor

37

Management functions



5 小结

- 以数据为核心，阐述了信息类服务系统的过程和方法。
- 以语义网和知识图谱为核心，讨论了海量信息服务的语义管理方法。
- 从信息服务类低代码平台出发，阐述了信息类服务软件的抽象和实现方法。